- Plug in your ESP32 to your computer
- Open Arduino IDE
- Download the necessary board if you havent
- Select: ESP32 Wrover Module from boards and select the appropriate COM port from your device
- Upload this code:

```
/*
  Rui Santos & Sara Santos - Random Nerd Tutorials
  Complete project details at
https://RandomNerdTutorials.com/get-change-esp32-esp8266-mac-address
-arduino/
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included in all copies or substantial portions of the Software.
*/
#include <WiFi.h>
#include <esp_wifi.h>

void readMacAddress(){
  uint8_t baseMac[6];
  esp_err_t ret = esp_wifi_get_mac(WIFI_IF_STA, baseMac);
  if (ret == ESP_OK) {
    Serial.printf("%02x:%02x:%02x:%02x:%02x:%02x\n",
                  baseMac[0], baseMac[1], baseMac[2],
                  baseMac[3], baseMac[4], baseMac[5]);
  } else {
    Serial.println("Failed to read MAC address");
  }
}

void setup(){
  Serial.begin(115200);

  WiFi.mode(WIFI_STA);
  WiFi.STA.begin();
```

```
  Serial.print("[DEFAULT] ESP32 Board MAC Address: ");
  readMacAddress();
}

void loop(){

}
```

- Compile using the check circle in the top left, and upload to the ESP32 using the arrow in the top left
- After the Output terminal displays: Hard resetting via RTS pin... the code has uploaded and you're good to go
- Open the Serial Monitor using Ctrl + Shift + M
- Press the EN button on the ESP32, this will display the devices MAC address down at the bottom of the Serial Monitor panel, it should look something like: [DEFAULT] ESP32 Board MAC Address: c8:2e:18:f7:4c:e8
- Once you have the MAC address, go to this site and follow the instructions, we're registering the ESP32 on the Utexas IOT network for wifi connection
- Click the link in the first step, sign in, then click Register WiFi-Device
- Input the MAC address that was printed in the Serial Monitor. For Device Description put your team name + ESP32
- Set Netowrk Profile to Unprotected and click Register
- You should see something like:

Registration completed for <MAC ADDRESS>

Network profile "Unprotected" will be applied to this device.

- When using *utexas* WiFi network, log in using your UT EID and UT EID password.
- When using *utexas-iot* WiFi network, log in using this PSK (no spaces): <PSK>
  - Now we're going to create a new sketch for the webserver, open a new sketch using Ctrl + N
  - Copy this code into the new sketch

```
#include <WiFi.h>


// Replace with your network credentials
const char* ssid     = "utexas-iot";
const char* password = "153260095736g35923140";


// Set web server port number
WiFiServer server(80);


const int controlPin = 2;


void setup() {
  Serial.begin(115200);


  // Set pin 2 as output
  pinMode(controlPin, OUTPUT);
  digitalWrite(controlPin, LOW); // Initially LOW


  // Connect to Wi-Fi
  Serial.print("Connecting to WiFi");
```

```
    WiFi.begin(ssid, password);

    while (WiFi.status() != WL_CONNECTED) {
      delay(1000);
      Serial.print(".");
    }

    Serial.println("");
    Serial.println("WiFi connected.");
    Serial.println("IP address: ");
    Serial.println(WiFi.localIP());

    // Start the server
    server.begin();
  }

  void loop(){
    // Check if a client has connected
    WiFiClient client = server.available();

    if (client) {
      Serial.println("New Client connected.");
      String currentLine = "";

      // Read client request
      while (client.connected()) {
        if (client.available()) {
          char c = client.read();
```

```
          Serial.write(c);
-
-
          if (c == '\n') {
-           // Handle keypress events
-           if (currentLine.indexOf("GET /W_ON") >= 0) {
-             digitalWrite(controlPin, HIGH);
-             Serial.println("Pin 2 set to HIGH");
-           }
-           else if (currentLine.indexOf("GET /W_OFF") >= 0) {
-             digitalWrite(controlPin, LOW);
-             Serial.println("Pin 2 set to LOW");
-           }
-
-           // Send a simple HTTP response
-           client.println("HTTP/1.1 200 OK");
-           client.println("Content-type:text/html");
-           client.println();
-           client.println("<html><body><h1>ESP32 Web Server</h1>");
-           client.println("<script>");
-           client.println("document.addEventListener('keydown',
  function(event) {");
-           client.println("  if (event.key === 'w' || event.key ===
  'W') {");
-           client.println("    fetch('/W_ON');");
-           client.println("  }");
-           client.println("});");
-           client.println("document.addEventListener('keyup',
  function(event) {");
```

```
            client.println("  if (event.key === 'w' || event.key ===
  'W') {");
            client.println("    fetch('/W_OFF');");
            client.println("  }");
            client.println("});");
            client.println("</script>");
            client.println("</body></html>");
            client.println();
            break;
          } else if (c != '\r') {
            currentLine += c;
          }
        }
      }
      // Close the connection
      client.stop();
      Serial.println("Client disconnected.");
    }
  }
```

- This will make it so every time you press W on the keyboard while in the webserver, the GPIO Pin 2 will turn on, triggering the blue LED light
- Make sure you replace the password with the PSK you got in the UT Network Portal, remove the spaces
- Compile the code and upload it to the ESP32
- Open the Serial Monitor and you should see `Connecting to WiFi....`
- After it connects you will have to scroll down to see your ESP's IP address
- Copy and paste the into the web browser and it should connect ot a web page

- Press W and watch the LED light up!!!